

0.0.1 Simulation Procedure

To determine the performance of EGARCH models under the five innovation distributions, a Monte Carlo simulation experiment will be conducted. In the simulation process, $R = 5000$ replications and four different sample sizes, $T = 50, 500, 1000$, and 2000 , will be considered. During the data generation process, five innovation distributions will be specified within the EGARCH model, namely the Normal, Student's t , Generalized Error Distribution (GED), Skewed Student's t , and Skewed Generalized Error Distribution (SGED).

The simulation procedure is described as follows:

- i. Specify a univariate EGARCH(1,1) model given by

$$r_t = \mu + \varepsilon_t, \quad (1)$$

$$\varepsilon_t = z_t \sqrt{h_t}, \quad (2)$$

$$\log(h_t) = \omega + \beta \log(h_{t-1}) + \alpha (|z_{t-1}| - E|z_{t-1}|) + \gamma z_{t-1}, \quad (3)$$

where r_t denotes the return series, ε_t is the innovation term, h_t is the conditional variance, and z_t is the standardized innovation process. The parameter ω is the constant term in the variance equation, α measures the magnitude effect of shocks, β captures volatility persistence, and γ measures the asymmetric or leverage effect.

- ii. Set the EGARCH(1,1) parameter values as

$$\omega = -0.10, \quad \alpha = 0.15, \quad \beta = 0.90, \quad \gamma = 0.10.$$

These parameter values are adopted from a recent asymmetric GARCH simulation study, where EGARCH(1,1) and GJR-GARCH(1,1) models were simulated using $\omega = -0.1$, $\alpha = 0.15$, $\gamma = 0.1$, and $\beta = 0.9$.

- iii. Specify the innovation distribution parameters as follows:

$$z_t \sim N(0, 1),$$

for the Normal innovation;

$$z_t \sim t_\nu, \quad \nu = 8,$$

for the Student's t innovation;

$$z_t \sim GED(k), \quad k = 1.5,$$

for the Generalized Error Distribution;

$$z_t \sim SSTD(\nu, \lambda), \quad \nu = 8, \quad \lambda = -0.5,$$

for the Skewed Student's t distribution; and

$$z_t \sim SGED(k, \phi), \quad k = 1.5, \quad \phi = -0.5,$$

for the Skewed Generalized Error Distribution.

The values $\nu = 8$ for the Student's t distribution and $k = 1.5$ for the GED are adopted from the asymmetric GARCH simulation study of Okereke et al., while the skewness values $\lambda = -0.5$ and $\phi = -0.5$ are selected to introduce moderate left-skewness into the skewed innovation distributions.

- iv. Using the specification in steps (i)–(iii), simulate return series from the EGARCH(1,1) process under each of the five innovation distributions. The innovation distribution will be alternated among the Normal, Student's t , GED, Skewed Student's t , and SGED distributions.
- v. Fit and estimate EGARCH(1,1) models to the simulated return series under each of the five competing innovation distributions: Normal, Student's t , GED, Skewed Student's t , and SGED.
- vi. Repeat the simulation procedure described in steps (i)–(v) for $R = 5000$ replications and for each sample size $T = 50, 500, 1000$, and 2000 . For each fitted model, compute model adequacy and forecasting performance measures, including the Akaike Information Criterion (AIC), Bayesian Information Criterion (BIC), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE). The average values of these measures across all replications will be used to compare the performance of the competing innovation distributions.